

Steffen Frey

Born 22. November 1982

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Academic Career

- since 2026 **Associate Professor**, *University of Groningen, Scientific Visualization and Computer Graphics (SVCG)*
- 2020 – 2025 **Assistant Professor**, *University of Groningen, Scientific Visualization and Computer Graphics (SVCG)*
- 2014 – 2020 **Postdoctoral Researcher**, *University of Stuttgart, Visualization Research Center (VISUS)*
Visionary Postdoc, *Cluster of Excellence “Data-integrated Simulation Science” (SimTech)*
- 2008 – 2014 **Dr. rer. nat.**, *Stuttgart, 07.11.2014 (summa cum laude)*, “Strategies for Efficient Parallel Visualization”, Supervisor: T. Ertl, GS SimTech
- 2010 & 12 **Visiting Researcher**, *University of California, Davis, USA*, Prof. Kwan-Liu Ma, VIDI Labs
- 2005 – 2006 **Study Abroad**, *University of Kansas, Lawrence, USA*, with Science Scholarship
- 2002 – 2008 **Dipl.-Inf.**, *University of Stuttgart, Computer Science*, thesis “GPU-based Cone Beam Reconstruction of Large CT Datasets” in collaboration with Daimler AG

Awards (see appendix for conference abbreviations)

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| 2023 | Best Paper , <i>EGPGV</i> [1] | 2019 | Winner , <i>VIS SciVis Contest</i> [4] |
| 2023 | Teacher of the Year 22/23 (CS) , Runner up | 2019 | Honorable Mention , <i>ISAV</i> [6] |
| 2022 | Best Paper , <i>EG (Education track)</i> [2] | 2019 | Best Paper , <i>EuroVis (Short)</i> [5] |
| 2020 | Best Paper , <i>LDAV</i> [3] | 2018 | Best Paper , <i>IV</i> [7] |

Funding / Principal Investigator (PI)

- 2017 & 21 & 25 **Visualization of Multi-field Processes in Porous Media (SFB 1313)**, *German Research Foundation (DFG); project D01 within the Collaborative Research Center 1313 “Interface-Driven Multi-Field Processes in Porous Media — Flow, Transport and Deformation”*. Three funding periods (FP) for a 4-year PhD position and hardware, jointly awarded to me and partners in Stuttgart: FP 1 (2017): T. Ertl, €268700; FP 2 (2021): T. Ertl & G. Reina, €285600; FP 3 (2025): G. Reina, €326400
- 2024 **Perception-action integration in brain-computer interfaces, robotic task planning, and adaptive visualization (FSE)**, *FSE Research Grant*, two 4-year PhD positions (€375000), with co-PIs A. Sburlea and H. Kazai, Note: this successful proposal builds in part on my NWO VIDI Science 2022 submission “Cognitive Model-based Optimization of Visualization for Large Scientific Data”, which was rejected after the interview stage (rating: Very Good)
- 2024 **Visual Data Science to Master Complex Simulation Ensembles (SimTech)**, *DFG; project in the SimTech Cluster of Excellence*, 2-year PhD student position, collaborative applicant with H. Hauser and M. Fjeld; lead applicant: M. Sedlmair
- 2019 **Machine Learning for Data-driven Visualization (SimTech)**, *DFG; project in SimTech*, 2-year PhD student position, collaborative applicant with I. Steinwart; lead applicant: T. Ertl
- 2019 **Quantifying Visual Computing Systems (SFB TRR 161)**, *DFG; project A02 in the Transregional Collaborative Research Center 161 “Quantitative Methods for Visual Computing”*, FP 2, 4-year PhD student position, €307100, co-PI with T. Ertl
- 2017 **Model-based Visual Analysis of Large Spatio-Temporal Data**, *DFG; project in SimTech*, 2-year PhD student position, co-PI with T. Ertl

Funding / CSC PhD positions China Scholarship Council

- 2021,23-25 **supported successful proposals of Z. Yin, Y. Xu, J. Wang, C. Li**

PhD Students / Co-Supervisor

- since 2026 **T. Daniel**, *Visualization of Multi-field Processes in Porous Media*, DFG, double-degree planned–University of Stuttgart
- since 2025 **D. Boerema**, *Neural Task Planning for Optimizing Visualization and Robot Interaction*, FSE

- since 2025 **O. Çağrı**, *Context-specific Grasping Control and Adaptive Visual Interfaces*, FSE
- since 2025 **C. Li**, *Multi-modal Medical Image Analysis and Visualization*, CSC, double-degree–U. of Science & Technology Beijing (USTB)
- since 2024 **J. Wang**, *Data- and model-driven differentiable fluid simulation*, CSC, double-degree–USTB
- since 2022 **Z. Yin**, *3D Multimodal Registration and Visualization for Augmented Reality Guided Surgery*, CSC
- 2022-25 **Y. Xu**, *Smoothed Particle Hydrodynamics Simulation in Computer Graphics*, CSC, Defense: 21th January 2025, double-degree–USTB
- 2021-25 **H. Gadirov**, *Automatic Configuration of Scientific Visualization Systems using Optimization and Machine Learning*, Defense: 14th October 2025

PhD Students / Supervision Involvement

- 2020-25 **H. Tärner (Duisburg-Essen)**, *Interactive Evaluation and Contextualization of Software Performance*
- 2020-24 **A. Gupta (Dresden & MPI-CBG)**, *Interactive In Situ Visualization of Large Volume Data*
- 2021-25 **A. Straub (Stuttgart)**, *Visualization of Multiphase Flow at Interfaces*, SFB 1313
- 2018-19,25 **M. Heinemann (Stuttgart)**, *Interactive Visualization of Droplet Dynamic Processes*, SFB 1313
- 2017-22 **G. Tkachev (Stuttgart)**, *Prediction and Similarity Models for Visual Analysis of Spatiotemporal Data*, SimTech
- 2016-21 **V. Bruder (Stuttgart)**, *Performance quantification of visualization systems*, SFB TRR 161

Service / Conference & Journal Organization

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| since 2025 LDAV , Steering Committee | 2021 LDAV , Poster Chair |
| since 2024 JDSSV , Associate Editor | 2021 PacificVis , Poster Co-Chair |
| since 2024 LDAV , Symposium Chair | since 2020 EGPGV , Steering Committee |
| since 2023 EGPGV , Chair of Steering Committee | 2020 EGPGV , Symposium Chair |
| 2023 BigVis , Paper Chair | 2019 EGPGV , Program Co-Chair |
| 2022 & 23 LDAV , Paper Chair | 2018 QiVC , Poster Chair |
| 2022 & 23 PacificVis , VisNotes Co-Chair | since 2016 WOIV , Organizer |

Service / Program Committee

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| 2025 VIS (Uncertainty Workshop) | 2019-22 VINCI | 2017-20 EuroVis (Short Papers) |
| 2021 & 25 VIS (Short Papers) | 2018 & 23 SC | 2017-18 ISC |
| since 2020 VisAI | 2018-23 SEAAA | 2016-21 ISVC |
| since 2019 SIBGRAPI | 2018-2021 ISAV | 2016-21 LDAV |
| since 2018 VIS | 2018-21 VMV | 2016 & 17 SA Asia Vis |
| 2020 & 22 ICPR | 2018-20 INFOCOMP | |
| 2020 BigVis | 2018 SC Asia | |

Invitations / Presentations

- 2026 **DSSV**, TBD
- 2024 **DEEA**, *Visualization of Large Spatio-Temporal Data Collections*
- 2024 **RUG Applied AI Symposium**, *Machine Learning for Large Data Visualization*
- 2024 **RUG Sem. Computational & Numerical Mathematics**, *Methods and Design Approaches for Fluid Dynamics Vis.*
- 2023 **TU Eindhoven Visualization Seminar**, *Scientific Visualization from an Optimization Perspective*
- 2022 **BigVis**, *Visual Mapping, Comparison and Exploration of Large Multifield Data*
- 2021 **Pretty Porous Lecture Series (SFB 1313)**, *Visualization of Multifield Data — Layouts, Features, and Systems*
- 2017 **NVIDIA GTC**, *Fast Flow-based Distance Quantification and Interpolation for High-Resolution Density Distributions*
- 2017 **China-Germany Visualization Workshop**, *High Performance Visualization of Volume and Time Series Data*

- 2015 **ISC (Workshop)**, *Reduced Representations for In-Situ Visualization*
- 2014 **GI BDVC**, *Quantifying Visual Computing Systems*
- 2013 **ParCo (Workshop)**, *Parallel Interactive Visualization: Strategies and Examples*
- 2009 **NVIDIA GTC**, *Memory Saving Fourier Transform on GPUs*

Invitations / Other

- 2025 **Dagstuhl Seminar**, *From Psychology Enabled Visualization to Visualization Enabled Psychology*
- 2022 **Dagstuhl Seminar**, *Anticipatory Human-Machine Interaction*
- 2018 **Dagstuhl Seminar**, *In Situ Visualization for Computational Science*
- 2012 **High Performance Visualization: Enabling Extreme-Scale Scientific Insight**, *book chapter*

Outreach

- 2025 **European Researchers' Night**, *Interactive Demonstration*, Presentation of VRT and WRT
- 2024 **Periodiek**, *Article*, Tursun, C., **Frey, S.**, Kosinka, J., "On Foveated Rendering," *Periodiek*, no. 1, 2024
- 2024 **Zpannend Zernike**, *Interactive Demonstration*, Presentation of Virtual Volume Raycaster (WRT)
- 2023 & 24 **European Researchers' Night**, *Interactive Demonstration*, Virtual Ray Tracer (VRT)
- 2022 & 23 **Zpannend Zernike**, *Interactive Demonstration*, VRT
- 2022 **Zpannend Zernike**, *Interactive Demonstration*, Webtool for heat data visualization
- 2020 **"Pretty Porous–Alles Porös" Science Exhibition**, *Interactive Demonstrator*, Planetarium Stuttgart, 18.06.–30.08.

Appendix: Conference & Journal Abbreviations

Abbreviation	Full Name
BigVis	Workshop on Big Data Visual Exploration and Analytics
DSSV	Data Science, Statistics, and Visualization
EuroVis	Eurographics/IEEE Symposium on Visualization
EG	Conference of the European Association for Computer Graphics
EGPGV	Eurographics Symposium on Parallel Graphics and Visualization
ICPR	International Conference on Pattern Recognition
INFOCOMP	International Conference on Information and Computing Science
ISC	International Supercomputing Conference
ISAV	In Situ Infrastructures for Enabling Extreme-Scale Analysis and Visualization
ISVC	International Symposium on Visual Computing
IV	International Conference on Information Visualisation
JDSSV	Journal of Data Science, Statistics, and Visualization
LDAV	IEEE Symposium on Large Data Analysis and Visualization
PacificVis	IEEE Pacific Visualization Symposium
QiVC	International Conference on Quantification in Visual Computing
SA Asia Vis	SIGGRAPH Asia Visualization / Asia Visualization Workshop
SC	International Conference for High Performance Computing, Networking, Storage and Analysis
SC Asia	Supercomputing Asia Conference
SEAAA	EUROMICRO Conference on Software Engineering and Advanced Applications
SIBGRAPI	Conference on Graphics, Patterns and Images
VIS	IEEE Visualization Conference
VisAI	Workshop on Visualization for AI Explainability
VINCI	International Symposium on Visual Information Communication and Interaction
VMV	Vision, Modeling, and Visualization
WOIV	International Workshop on In Situ Visualization